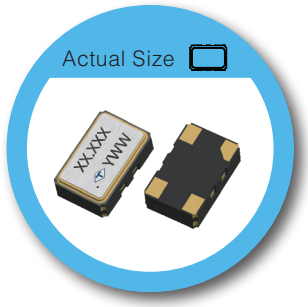
TAITIEN
Our Performance, Your Reputation

TW-F Type Ultra Wide Temperature 5.0 x 3.2 mm SMD Voltage Controlled Temperature Compensated Crystal Oscillator



RoHS Compliant

FEATURE

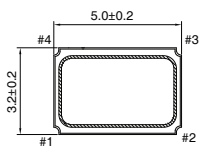
- Typical 5.0 x 3.2 x 1.85 mm ceramic SMD package.
- Ultra High Temperature for -55°C ~ +105°C, ±0.2ppm, -55°C ~ +125°C, ±0.5ppm
- Frequency support from 10MHz to 52MHz
- Wide temperature range
- Pb-free/RoHS compliant

TYPICAL APPLICATION

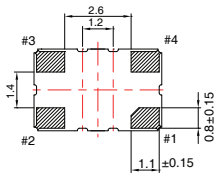
- Aerospace
- Industrial Equipment
- Base Stations
- Energy and Power

DIMENSION (mm)

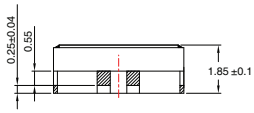
[TOP VIEW]



[BOTTOM VIEW]

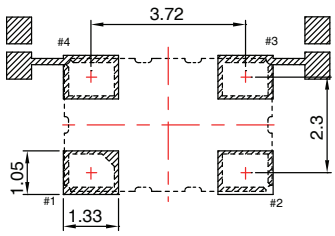


[SIDE VIEW]



Pin#	Function
1	Vcon:VC-TCXO Floating:TCXO
2	GND
3	OUTPUT
4	V_{DD}

SOLDER PAD LAYOUT (mm)



To ensure optimal oscillator performance, place a by-pass capacitor of 0.1μF as close to the part as possible between Vdd and GND pads.

ELECTRICAL SPECIFICATION

Parameter			3.3V Min.	3.3V Typ.	3.3V Max.	Unit	Test Condition
Supply Voltage Variation (V_{DD})			V_{DD} – 5%	V_{DD}	V_{DD} + 5%	V	
Frequency Range			10	-	52	MHz	
Frequency Tolerance (25°C)			-	-	±1.0	ppm	
Frequency Stability	vs Supply Voltage (±5%) Change		-	-	±0.1	ppm	
	vs Load (±10%) Change		-	-	±0.1	ppm	
	vs Aging (@1ST Year)		-	-	±1.0	ppm	
Supply Current	CMOS	10MHz ≦ F_0 ≦ 38MHz	-	-	5.0	mA	
		38MHz < F_0 ≦ 52MHz	-	-	8.0	mA	
	Clipped Sine Wave	10MHz ≦ F_0 ≦ 38MHz	-	-	4.5	mA	
		38MHz < F_0 ≦ 52MHz	-	-	5.0	mA	
Output Level	CMOS	Output High	90%V_{DD}	-	-	V	
		Output Low	-	-	10%V_{DD}	V	
	Clipped Sine Wave		0.8	-	-	V_{P-P}	
Load	CMOS		-	-	15	pF	
	Clipped Sine Wave		-	10 KΩ // 10pF	-		
Transition Time (10% ~ 90%)		Rise Time	-	-	3.0	nSec	
		Fall Time	-	-	3.0	nSec	
Duty Cycle			45	-	55	%	
Tri-State Control	Enable		80%V_{DD}	-	-	V	
	Disable		-	-	20% V_{DD}	V	
Start Time			-	-	5	mSec	
Control Voltage Range (VCTCXO)			0.5	-	2.5	V	
Pulling Range (VCTCXO)			±5.0	-	-	ppm	
Vc Input Impedance (VCTCXO)			100	-	-	KΩ	
Phase Noise @ TCXO V_{DD} =3.3V, Fount=10MHz Please add filter capacitor as suggested, and 33nF at pin7		100Hz offset	-	-116	-	dBc/Hz	
		1KHz offset	-	-140	-	dBc/Hz	
		10KHz offset	-	-147	-	dBc/Hz	
		100KHz offset	-	-150	-	dBc/Hz	

Note: not all combination of options are available. Other specifications may be available upon request.

FREQ.STABILITY vs. TEMP. RANGE

ppm / Temp.(°C)	±0.2	±0.28	±0.5
-55 ~ +95	△	○	○
-55 ~ +105	△	○	○
-55 ~ +125	X	X	○

* ○: Available △:Conditional X: Not available

Note: not all combination of options are available. Other specifications may be available upon request.

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